

US Payments & Transactions Market: Strategic Analysis

Senior Strategy Consultant Perspective | March 2026

EXECUTIVE SUMMARY

The US payments market is undergoing the most structurally significant transformation in a generation. Three forces are converging simultaneously: (1) **real-time rail proliferation** (FedNow, RTP) threatening card economics at the margin; (2) **stablecoin legitimization** via the GENIUS Act (July 2025), opening a new settlement layer that could bypass correspondent banking entirely; and (3) **AI-driven intelligence** shifting competitive advantage from infrastructure ownership toward data density and model quality. The result is a market where the traditional integrated stack — bank + network + processor + issuer — is fragmenting horizontally, with software and intelligence layers capturing an increasing share of value. Yet cards remain structurally dominant in consumer payments, and incumbents with scale, data, and distribution are not standing still. The next 3–5 years will determine which players become utilities and which become intelligence control points — and the strategic choices made now will define the industry structure for the following decade.

1. MARKET STRUCTURE AND VALUE POOLS

The End-to-End US Payments Value Chain

The US payments system is best understood as a set of overlapping value chains, not a single stack. Each payment type has its own rails, economics, and players — but AI, APIs, and regulatory shifts are forcing convergence.

Layer 1 — Issuers: Banks and credit unions originating cards and accounts. Revenue: interchange, interest income, annual fees, FX spreads on international cards. Key players: JPMorgan Chase, Citibank, Bank of America, Capital One, American Express (issuer/network hybrid), Synchrony Financial, Bread Financial. Issuer economics are heavily reliant on interchange (\$100B+ annually in US credit and debit interchange revenue) and revolving credit spread.

Layer 2 — Card Networks: Visa, Mastercard (four-party model); Amex, Discover (three-party model). Revenue: network/assessment fees (~0.11–0.13% per transaction), data services, value-added services (fraud tools, analytics, tokenization). In fiscal year 2025, Visa processed \$14 trillion in total payments volume across 258 billion transactions, generating \$40 billion in net revenue. Mastercard handled \$9.2 trillion in GDV. Their moats are ubiquity, token vaults, fraud

infrastructure, and rule-setting authority — not transaction processing per se. These are the strongest structural positions in global payments.

Layer 3 — Acquirers and Processors: Merchant-facing. Revenue: acquiring spread, gateway fees, value-added services. Collectively, merchant acquirers handle more than \$9 trillion in annual transaction volume in the US. Key players: FIS (Worldpay), Fiserv (Clover), Global Payments, JPMorgan Merchant Services, Stripe, Adyen, Block/Square. Commoditization pressure is acute here, especially in gateway and plain-vanilla processing.

Layer 4 — Gateways and Orchestration: Connecting merchants to acquirers, routing transactions. Standalone gateway revenue is compressing toward zero as orchestration layers absorb the function. Players: Spreedly, Primer, Gr4vy, Checkout.com, Stripe (vertically integrated). Orchestration is the fastest-growing layer — merchants want multi-acquirer routing, least-cost routing, and authorization rate optimization simultaneously.

Layer 5 — Core Banking and Treasury Infrastructure: BaaS platforms (Galileo/SoFi, Marqeta, Unit, Synctera), core banking (FIS, Fiserv, Jack Henry), treasury infrastructure (Kyriba, Modern Treasury, TreasuryPrime). These are the rails under the fintechs. Revenue: SaaS fees, per-transaction fees, float on held balances.

Layer 6 — Correspondent Banks and Wholesale Infrastructure: SWIFT messaging, FedWire, CHIPS for large-value settlement, correspondent networks for cross-border. Revenue: wire fees, FX spreads, float, correspondent account balances. Under structural pressure from RTP, FedNow, and stablecoins.

Layer 7 — Wallets and Super-Apps: Apple Pay, Google Pay, PayPal/Venmo, Cash App, WeChat Pay (limited US). Revenue models vary: Apple earns ~15bps per Apple Pay transaction on credit, PayPal earns transaction take-rates of ~1.5-2%. Wallets are becoming the consumer identity layer, not just a payment credential holder.

Layer 8 — Crypto and Stablecoin Infrastructure: Circle (USDC), Paxos (PYUSD), Coinbase, Fireblocks (custody/settlement rails), Ripple (ODL for cross-border). Revenue: issuance float on stablecoin reserves (T-bill yield on ~\$50-100B+ USD-pegged reserves), custody fees, exchange spreads, payment network fees. The volume of outstanding USD stablecoins grew to nearly \$280 billion at year-end 2025 from about \$25 billion in 2020.

Layer 9 — Fraud, Risk, and Compliance Infrastructure: Featurespace, Sardine, Sift, Socure, Alloy, Persona, Onfido, ComplyAdvantage, unit21, Hawk AI. Revenue: SaaS, per-API-call, per-transaction risk scoring. This layer is growing faster than any other as AI shifts it from rules-based to model-driven.

Revenue Pool Sizing and Concentration

Revenue Pool	Estimated US Annual Size	Where Profit Concentrates
Credit card interchange	~\$80-90B	Card issuers (JPM, Citi, Amex)
Debit card interchange	~\$25-30B	Issuers (regulated by Durbin)
Network/scheme fees	~\$15-20B	Visa, Mastercard
Acquiring/gateway margins	~\$12-18B	Large acquirers, Stripe, Adyen
Cross-border FX spread	~\$15-25B	Banks, Amex, Visa, specialized fintechs
Wire/SWIFT fee income	~\$5-8B	Large correspondent banks
Float income (wallets, BaaS)	~\$3-6B	PayPal, Cash App, neobanks
Fraud/risk services	~\$4-7B	Networks, fintechs, incumbents
AML/KYC/compliance SaaS	~\$3-5B	Fintechs, large bank-owned platforms
Stablecoin reserve float	~\$12-16B	Circle, Paxos, bank issuers

Key insight: The single largest profit pool in US payments is credit card interchange — controlled by issuers and defended by the network-issuer incentive structure. It is both the most economically significant pool and the most politically/regulatorily vulnerable. The second structural insight is that stablecoin reserve float is already material and could rival wire fee income within 3 years.

Where Value Is Being Created vs. Defended vs. Disrupted

- **Created:** AI-powered risk scoring, payment orchestration, stablecoin reserve float, embedded payments SaaS
- **Defended:** Card network ubiquity (tokenization moats), issuer rewards economics, JPMorgan's cross-border infrastructure
- **Disrupted:** Acquiring margins (race to zero), cross-border FX spread (Wise, stablecoins), wire fees (FedNow, RTP), check processing (secular decline)

2. PAYMENT TYPES: STRATEGIC ANALYSIS

Cards (Credit, Debit, Prepaid — Consumer and Commercial)

Current role: Still the dominant non-cash payment method. Credit cards dominated in-store transactions at 70% in 2024. US consumers hold ~170 million credit cards; credit card debt exceeded \$900B in 2023. Cards are the default payment UX for e-commerce, in-store, and recurring billing.

Economics: Consumer credit: interchange 1.5–2.4% + flat fee. Debit (unregulated): ~1.5% interchange. Debit (regulated, Durbin): capped at ~0.21% + \$0.05. Commercial/corporate cards: 2.0–2.8%, higher interchange due to card-not-present and B2B premiums. Merchant discount rate aggregate ~1.8–2.5% in most verticals, higher for premium and travel cards. A standard US consumer credit card transaction might carry an interchange rate of 1.50% + \$0.10, while a premium rewards card could reach 2.40% + \$0.10 or higher.

Key players: Visa, Mastercard (networks); JPMorgan Chase, Amex, Citi, Capital One, BofA (top issuers); Fiserv, FIS, Adyen, Stripe (acquiring side). Marqeta leads card issuing infrastructure for fintechs.

Strengths: Global acceptance ubiquity. Consumer protection (chargebacks, zero liability). Rewards and benefits economics that are self-reinforcing. Network effects on both merchant and consumer sides. Tokenization infrastructure is now deeply embedded in mobile wallets (Apple Pay = card credentials, not a new rail).

Weaknesses: High cost to merchants, especially SMBs. Interchange attracts regulatory scrutiny. Credit Card Competition Act (CCCA) — if passed, would force routing choice on credit cards (analogous to Durbin for debit), threatening the Visa/Mastercard duopoly on credit card routing. Chargebacks/disputes are structurally expensive for issuers and merchants alike.

Strategic outlook (3–5 years): Cards will remain dominant in consumer payments by volume and value, but face two structural pressures: (1) incremental margin compression on interchange from regulation and competition; (2) selective displacement by A2A/RTP in specific merchant categories (utilities, rent, B2B). The networks themselves are well-positioned: Visa and Mastercard's response to A2A is tokenization of accounts (Visa A2A), not resistance. Commercial card is the fastest-growing sub-segment (expense management, AP automation, virtual cards). Prepaid and BaaS-linked debit will grow as embedded finance platforms issue white-label debit products.

ACH (Automated Clearing House)

Current role: The backbone of B2B and recurring consumer payments. Payroll, direct deposit, bill pay, B2B supplier payments. Not real-time — next-day and same-day ACH are available but not instant.

Economics: Very low per-transaction cost. Same-Day ACH has a network fee of \$0.052 per transaction. Volume: ~31 billion transactions and ~\$80 trillion in value annually (Nacha 2024).

Per-transaction revenue is minimal — value capture is in treasury, float, and volume-driven SaaS.

Key players: Federal Reserve (FedACH), EPN (operated by The Clearing House), Nacha (governance). Access via bank treasuries, fintechs using Modern Treasury, Dwolla, Plaid, or direct bank APIs.

Strengths: Near-universal bank coverage. Very low cost. Deeply embedded in payroll, direct deposit, bill pay, B2B AP workflows.

Weaknesses: Not real-time (even Same-Day ACH settles in batches). Not irrevocable — returns/reversals create fraud risk and operational complexity. Not 24/7. Poorly suited to high-urgency or consumer-demand use cases.

Strategic outlook: ACH will retain dominance in high-value, non-urgent B2B flows for the next 5+ years. Same-Day ACH will grow but will increasingly be complemented (not replaced) by FedNow and RTP for time-sensitive use cases. The key risk: as FedNow reaches critical mass, merchants and platforms may shift recurring consumer billing to A2A instant rails for lower cost and faster float benefit.

RTP / FedNow / Instant A2A Payments

Current role: The most strategically important infrastructure shift in US payments in decades. RTP (The Clearing House, launched 2017, now covering ~90% of US demand deposit accounts) and FedNow (launched July 2023) are competing 24/7/365 instant payment rails. FedNow expanded from 35 launch banks to more than 900 institutions by August 2024, with community banks representing 78% of participants.

Economics: Network fees very low — \$0.045 per transaction on FedNow. Revenue captures is in treasury management, float benefit to receivers (immediate funds availability), reduced chargeback exposure, and the ability to build new products on top (e.g., request-for-payment, insurance claims, gig worker payouts, earned wage access).

Key players: Federal Reserve (FedNow), The Clearing House (RTP), bank participants, fintech overlays (Dwolla, Modern Treasury, Plaid), and software orchestrators building on top.

Strengths: Real-time, 24/7, irrevocable (no chargebacks). Removes float drag. Enables new payment flows that are impossible on ACH or card rails. Lower cost than cards for merchants willing to accept the UX change.

Weaknesses: Irrevocability is a double-edged sword — no consumer protection equivalent to card chargebacks. Authorized push payment (APP) fraud is a growing problem (significant in UK after Faster Payments launch). Merchant acceptance requires new checkout UX; consumer adoption is a chicken-and-egg problem. Credit functionality (BNPL, installments) must be layered separately — no native credit.

Strategic outlook: The most important 3–5 year question in US payments. The consensus bullish case: RTP/FedNow displace check and ACH in bill pay, insurance payouts, gig/contractor payments, and B2B disbursements. The bearish case on card displacement: without consumer protection parity and a rewards equivalent, consumers won't voluntarily shift from credit cards for discretionary purchases. The honest answer: significant displacement in push-payment use cases (payroll, earned wage access, insurance, government disbursements, B2B AP) but limited consumer POS displacement through 2028 absent a regulatory forcing function or a merchant-funded incentive structure. **Non-obvious second-order effect:** FedNow + open banking + request-for-payment (RfP) protocols together could create a merchant-directed "pull A2A" product competitive with debit cards for subscription billing — this is the real threat to regulated debit interchange over the next 5 years.

Checks

Current role: In secular decline but still the largest payment instrument by dollar value in B2B. Cash usage was approximately 20% of all transactions in 2021, and by 2024 the percentage decreased to 16%. Checks follow a similar arc. Federal Reserve data shows checks declining ~7% per year. B2B still accounts for ~60% of check volume; consumer check writing is approaching residual.

Economics: Costly to process — bank revenue from check processing comes from service fees, float during clearing, and overdraft fees triggered by bad check timing. Positive Pay, Check Fraud Fraud services generate some incremental bank revenue. Total check clearing costs to the industry are substantial but declining as volumes fall.

Key players: Federal Reserve (check processing services), FIS, Fiserv, bank back-offices. Fintech challengers: Checkbook.io (digital check issuance), Paymentus (electronic bill pay replacing check receipt).

Strategic outlook: Structural decline accelerating. The remaining check volume is sticky because it is embedded in accounts payable/receivable software (QuickBooks, SAP, Oracle), legal/real estate settlements, and segments that have not yet digitized. Digital check issuance services (sending a PDF or printable check via email) are a transitional product, not a destination. Check processing will become a utility service managed at declining scale by the Fed and large banks. No strategic premium here — only cost management.

Domestic Wires (Fedwire, CHIPS)

Current role: High-value, same-day settlement for corporate treasury, real estate, securities settlement, and interbank funding. CHIPS (Clearing House Interbank Payments System) settles ~\$1.8T/day; Fedwire ~\$4T/day. Not a consumer product.

Economics: Per-wire fee income for banks: \$15–50 per outgoing wire (consumer), variable for wholesale. CHIPS is net-settled with significant netting efficiency. Total wire fee revenue is relatively modest (~\$5–8B) compared to the dollar volume, because unit economics are transaction fees on very large amounts.

Key players: Federal Reserve (Fedwire), The Clearing House (CHIPS), money-center banks (JPMorgan, Citi, BofA, Wells Fargo). Corporate treasury management systems connect clients to wire infrastructure.

Strategic outlook: Wires remain critical for large-value, time-sensitive institutional payments. They are not being displaced by FedNow (different value threshold, different risk profile). The real competitive threat: blockchain-based settlement (tokenized deposits, stablecoins) for institutional cross-border flows, where JPMorgan's Onyx/JPM Coin has demonstrated that banks themselves see the opportunity to disintermediate SWIFT-dependent correspondent chains using internal ledgers.

SWIFT and Correspondent Banking

Current role: The messaging infrastructure (not a rail itself) connecting ~11,000 financial institutions in 200+ countries. Used for international wire instructions. ISO 20022 migration is now in full deployment phase.

Economics: High-margin for correspondent banks holding nostro/vostro accounts — float, FX spread, wire fees, and crucially, the ability to cross-sell trade finance and treasury services to respondent banks. Total correspondent banking revenue globally: ~\$25–35B (wholesale). In the US, money-center banks (JPMorgan, Citi, BofA, Wells) are the dominant USD correspondents.

Key players: SWIFT (messaging), JPMorgan (dominant USD correspondent), Citi (global network), BofA, Wells Fargo. Challengers: Wise, Ripple (ODL), stablecoin rails.

Strengths: Deep integration into trade finance, documentary credit, securities settlement. ISO 20022 migration is adding rich data payloads that improve reconciliation and enable AI-based processing.

Weaknesses: Slow (1–3 day settlement in many corridors), opaque FX pricing, high per-transaction cost (\$25–50 for cross-border wires), multiple intermediary hops. De-risking (correspondent bank exits from high-risk corridors) is reducing coverage and increasing concentration risk.

Strategic outlook: SWIFT itself is adapting — ISO 20022, pre-validation APIs, SWIFT Go for smaller cross-border payments. But the structural pressure from Wise/Airwallex, stablecoin rails, and Visa/Mastercard B2B Connect is real. The correspondent banking profit pool will compress 20–35% over 5 years as fintech and blockchain-based alternatives capture the most price-

sensitive corridors. Large-value institutional flows stay on SWIFT/Fedwire/CHIPS. SMB and consumer remittance flows are the contested territory.

International / Cross-Border Payments

Current role: Cross-border payments market is worth over \$194 trillion and is forecast to reach \$320 trillion by 2032. US is the largest source of remittance flows globally. Cross-border B2B payments are the highest-fee, highest-friction segment — and the most disrupted.

Economics: Banks capture FX spread of 1.5–3% on retail cross-border payments, plus wire fees. Fintechs like Wise operate at 0.3–0.7% all-in for major corridors. The spread compression is massive — early Wise corridors saw 80%+ fee reductions vs. banks. For B2B: Airwallex, Nium, Ebury, and Part quote interbank rates plus small FX margin for SMB flows.

Key players: JPMorgan Payments, Citi Treasury & Trade Solutions (large corporate). Wise, Airwallex, Nium, Ebury, Currencycloud (acquired by Visa), Ripple, Remitly, Western Union (declining). Airwallex reached \$1 billion in annualized revenue in October 2025, processing more than \$130 billion in annualized TPV.

Strengths of challengers: Real exchange rates, transparent pricing, local payment network access in 50+ markets, API-first, fast settlement.

Strategic outlook: The most disrupted segment in payments over the next 5 years. Stablecoins are a potentially transformative force here — over 50% of companies surveyed expect to adopt stablecoins for cross-border payments in the next 6–12 months, expecting stablecoins to handle 5–10% of cross-border payments by 2030, equivalent to \$2.1–4.2 trillion according to EY estimates. The key constraint is last-mile fiat conversion and regulatory compliance in destination markets. Whoever solves programmatic FX conversion with stablecoin rails at scale wins the next era of cross-border B2B payments.

Digital Wallets and Mobile Payments

Current role: Digital wallets accounted for almost 50% of online transactions in 2023, and their usage is expected to surpass 54% by 2025. Apple Pay and Google Pay dominate contactless in-store in the US. PayPal dominates online checkout. Venmo/Cash App own P2P. Shop Pay (Shopify) is the fastest-growing checkout wallet in e-commerce.

Economics: Wallets are primarily credential storage layers on top of existing rails (mostly card, some ACH/A2A). Apple earns ~15bps per credit card transaction routed through Apple Pay from issuers. Google Pay monetizes less directly — data and ecosystem lock-in. PayPal earns 1.5–2% take rate on transactions. Cash App monetizes through Bitcoin, instant transfer fees, and Cash App Card interchange. Venmo monetizes via Pay with Venmo merchant fees.

Key players: Apple Pay (iPhone NFC dominance), Google Pay, PayPal, Venmo, Cash App, Samsung Pay, Shop Pay, Amazon Pay, Walmart Pay.

Strategic outlook: Wallets are becoming the primary consumer payments interface — not the rail. The strategic importance is in owning the checkout experience and consumer identity, which confers data leverage and the ability to route across underlying rails. Apple's NFC monopoly in iPhone (now being partially opened by EU regulation, with US implications likely) has been a moat. The most important long-term wallet dynamic: **whoever controls the checkout flow controls routing decisions** — this is why Visa and Mastercard are investing in tokenization and network-level wallet integrations rather than fighting wallets. Stablecoin-linked wallets (PayPal's PYUSD, Coinbase's Base network) represent a potential future where wallets settle on non-card rails.

Stablecoins

Current role: The most strategically important emerging payment infrastructure in the US. Outstanding USD stablecoins reached nearly \$280 billion at year-end 2025. The GENIUS Act (July 2025) created the first federal regulatory framework for payment stablecoins. The OCC conditionally granted national trust bank charters to Circle, Paxos, and three other nonbank financial firms in December 2025.

Economics: Issuer earns T-bill yield on reserve assets backing stablecoins. With \$280B outstanding and short-term T-bills yielding ~4.5%, this represents roughly \$12-13B in annual float income — a revenue pool that did not exist five years ago. Transaction fees on stablecoin rails are near-zero (gas fees on USDC/Base transactions are fractions of a cent). The economic model is: issue stablecoin at near-zero transaction cost, earn float on reserves, retain users in ecosystem for other services.

Key players: Circle (USDC — dominant in institutional and B2B), Paxos (PYUSD, USDP), Tether (USDT — offshore), Coinbase (USDC co-issuer, Base L2 network), Fireblocks (settlement rails for institutional stablecoin flows), Ripple (RLUSD for ODL), bank issuers under GENIUS Act.

Strengths: Near-instant settlement (seconds on L2 networks). Global reach without correspondent banking. Programmability (smart contracts for conditional payments, escrow, streaming payments). Float to the issuer (not the merchant or consumer). GENIUS Act provides regulatory clarity for the first time.

Weaknesses: No consumer protection equivalent to card chargebacks. Last-mile fiat conversion friction. Regulatory complexity in non-US jurisdictions. GENIUS Act prohibits interest/yield to holders — limiting consumer adoption incentive.

Strategic outlook: The most transformative 5-year bet in payments. Use cases accreting fastest: (1) cross-border B2B (supplier payments, treasury corridors); (2) institutional settlement (tokenized securities, repo); (3) gig economy worker payouts; (4) remittances. **Non-obvious**

insight: Banks issuing stablecoins under GENIUS Act can capture the float income that currently accrues to Circle — this is why JPMorgan (JPM Coin for institutional), Citi, and BofA are all moving aggressively into tokenized deposit and stablecoin issuance. The real question is not whether stablecoins grow — they will — but whether the economic gains from reserve float accrue to bank issuers or to crypto-native infrastructure providers.

Other Crypto Payment Mechanisms

Current role: Minimal for actual commerce payments. Bitcoin is primarily a store of value and speculative asset, not a payment instrument (high transaction costs on base layer, volatility). Lightning Network (Bitcoin L2) enables micropayments with low fees but adoption is limited. Ethereum smart contracts are the backbone of DeFi but not consumer payments. Ripple/XRP is used in specific institutional cross-border corridors.

Strategic outlook: Non-stablecoin crypto for payments remains a niche use case except in specific remittance corridors where recipients prefer crypto-native settlement (e.g., Latin America, Sub-Saharan Africa). The focus of institutional energy has shifted entirely to stablecoins and tokenized deposits. Bitcoin's role in payments may be permanently limited by volatility and the availability of superior alternatives (stablecoins).

3. KEY PLAYERS AND COMPETITIVE DYNAMICS

Traditional Incumbents

JPMorgan Chase: The most powerful player in US payments by any measure. Largest card issuer (~\$1.4T in card sales volume), dominant correspondent bank, owns Paymentech (large merchant acquirer), has deployed JPM Coin for institutional cross-border, and is investing aggressively in tokenized deposits. JPM's competitive advantage is the combination of consumer deposits, credit extension, merchant relationships, and institutional infrastructure — no competitor has all four. Greatest risk: if embedded finance platforms disintermediate the merchant relationship, JPM loses cross-sell leverage.

Visa / Mastercard: The most structurally resilient players in the US market. Moats: (1) network ubiquity — every merchant, every bank; (2) token vaults — 10B+ network tokens issued, making tokenized card credentials the default secure credential; (3) data — the only players with full visibility across issuer and acquirer side of transactions; (4) value-added services (fraud tools, analytics, Visa Direct / Mastercard Send for push payments) now represent ~35% of network revenue and growing. Key risk: CCCA (Credit Card Competition Act) routing mandates for credit cards, which would compress network fee economics by forcing network competition. As part of the Visa/Mastercard interchange settlement in 2024, Visa agreed to hold off on interchange increases for five years.

Fiserv / FIS / Global Payments: The processing oligopoly. Between them, they handle the majority of US merchant acquiring volume. Revenue is increasingly SaaS-oriented (Clover, WorldPay). Margin compression from Adyen, Stripe is a persistent threat at the upper end of the market. Their defense: scale, SaaS bundles (Clover = POS + software + payments), and SMB/community bank relationships that fintech competitors cannot easily replicate.

American Express: The most profitable card issuer/network by net income per card. Closed-loop model gives Amex unique data advantages and eliminates interchange passthrough. Amex's key strategic move: expansion into SMB and B2B cards, where interchange is higher and the rewards subsidy is more manageable. Risk: premium consumer reward cost inflation and merchant surcharge acceptance.

Fintech Challengers

Stripe: The most important private payments infrastructure company. Estimated \$1T+ in annual TPV, \$15B+ in revenue. Stripe's edge: (1) developer-first API experience that has become the default for SaaS and e-commerce; (2) Stripe Radar (AI fraud — cuts card testing attacks by 80% per company data); (3) Stripe Connect (marketplace/platform payments); (4) expanding into financial services (Stripe Capital, Stripe Treasury, Stripe Issuing). Risk: Adyen is winning the enterprise mid-market; Stripe's move upmarket is contested.

Adyen: The enterprise PSP challenger. Outperforming legacy processors on gross margin due to: direct acquiring (no third-party acquirer margin), single platform architecture, and lower total cost of ownership for large merchants. Adyen's "unified commerce" thesis — single platform for in-store and online — is winning enterprise retailers. Key limitation: less suited for SMB or long-tail merchants than Stripe.

Block/Square: Dominant in SMB POS and Cash App ecosystem. Cash App is a financial services super-app with 57M+ monthly actives. Square's Afterpay acquisition brought BNPL to the suite. The risk: Cash App's monetization is heavily dependent on Bitcoin trading spread and Instant Transfer fees — both under competitive pressure.

PayPal: The dominant online checkout wallet (3.5B+ PayPal buttons on the web). Revenue under pressure as Venmo monetization matures, PayPal faces competition from Shop Pay/Apple Pay in checkout, and the company navigates a strategic transition toward AI-driven personalized checkout. PYUSD (PayPal stablecoin) is a meaningful stablecoin bet — volume growing but mostly ecosystem-internal.

Plaid: The infrastructure layer for open banking in the US. Connects consumer bank accounts to ~8,000 fintech apps via API. Revenue: per-API-call fees for account verification, instant ACH, identity. Strategic position: the data broker/aggregator sitting between banks and fintechs. Risk: bank direct access APIs (Chase, Citi building their own connectivity APIs), and CFPB 1033

rulemaking mandating open banking data portability — which both enables and commoditizes Plaid's core service.

Modern Treasury: B2B payment operations infrastructure. Connects to ACH, Fedwire, RTP, FedNow via bank APIs with workflow automation for reconciliation, money movement, and treasury operations. Represents the "payments operating system" thesis — software sitting above the rails, not on them.

Marqeta: Card issuing infrastructure. Powers Cash App, Uber, DoorDash, Klarna, and hundreds of BaaS fintechs. Revenue: interchange split with issuing banks and per-transaction fees. Risk: Stripe Issuing, Lithic, Highnote as competitors; margin pressure as issuers negotiate harder.

Cross-Border Specialists

Wise: The gold standard for transparent, low-cost international transfers. \$100B+ in annual cross-border volume. Competitive advantage: mid-market FX rate transparency, proprietary direct bank integrations in 80+ markets (bypassing correspondent banks), and network effects in payment flow matching. Risk: Airwallex and neobanks replicating the value proposition with broader feature sets.

Airwallex: The fastest-growing cross-border payment infrastructure company. Airwallex reached \$1 billion in ARR in October 2025, now repositioning as an embedded finance platform — 60% of revenue comes from its API and products. Strategic shift from FX spread to embedded finance SaaS is raising take rates and reducing commoditization risk.

Nium: B2B payment infrastructure in 100+ countries. Particularly strong in Southeast Asia and embedded finance distribution. SWIFT alternative for specific corridors.

Ripple / ODL: Uses XRP as a bridge currency for cross-border settlement in specific corridors (primarily USD/MXN, USD/PHP). Niche but growing in markets where traditional correspondent banking is expensive or unreliable.

Wallet / Platform Ecosystems

The key insight: **wallets are becoming the consumer identity layer in payments, not just a credential holder.** Apple Pay, Google Pay, and PayPal each have 100M+ US users with stored payment credentials. The strategic importance: whoever controls the checkout UI controls (a) routing decisions, (b) consumer data, and (c) the upsell into BNPL, rewards, and financial services.

Shop Pay (Shopify): The fastest-growing checkout button in US e-commerce. Shopify's ~25% of US e-commerce GMV gives Shop Pay significant network effects. Risk: Shopify's merchant concentration and fee structure make it a competitor to payment processors, not just a wallet.

Amazon Pay: Primarily used for off-Amazon sites. Limited network effect outside Amazon ecosystem.

Crypto / Stablecoin Infrastructure Players

Circle: Dominant USDC issuer. The primary beneficiary of GENIUS Act regulatory clarity. Circle's moat: USDC is the most trusted, regulated, and institutionally-accepted USD stablecoin. Circle's challenge: it must build distribution and use case adoption faster than banks (who now can issue their own stablecoins) replicate the trust and compliance infrastructure.

Fireblocks: Institutional-grade digital asset custody and settlement infrastructure. The "plumbing" for institutional stablecoin flows. Used by 1,800+ financial institutions. The SWIFT of blockchain settlement.

Coinbase: Consumer and institutional crypto exchange plus Base L2 network. USDC co-issuance with Circle. Coinbase's stablecoin ambition is to make Base the settlement layer for stablecoin commerce — low-cost, fast, programmable.

Paxos: Issuer of PYUSD (PayPal's stablecoin) and USDP. National trust bank charter (OCC) received December 2025. Regulated, compliant, enterprise-focused.

AI-Native Risk and Compliance Players

Sardine: Fraud and compliance API infrastructure. AI-first, covers device fingerprinting, behavior biometrics, and AML graph analytics in one platform. Backed by a16z.

Socure: Identity verification and fraud prediction. 98%+ auto-approval rates for consumer identity, significantly outperforming legacy ID verification. Embedded in fintech onboarding flows.

Sift: E-commerce fraud prevention. ML-driven. Strong in marketplace and platform fraud.

Alloy: KYC/KYB decisioning orchestration. API connects to 180+ data sources and runs compliance workflows for fintech and bank partners.

Persona: Identity verification SaaS. Modular, no-code compliance workflow builder. Fastest-growing KYC infrastructure for fintechs.

ComplyAdvantage: AI-driven AML and sanctions screening. Replaces legacy Accelus/Lexis-Nexis watchlist matching with a continuously updated AI model.

Unit21: Case management and AML/fraud investigation workflows. Replacing legacy FIS/Oracle FCRM systems.

4. TECHNOLOGY AND INFRASTRUCTURE SHIFTS

Real-Time Rails and 24/7 Settlement

The transition from batch (ACH, checks) to real-time (RTP, FedNow) is not a 5-year project — it is structurally underway. The challenge: real-time is irrevocable, which means fraud prevention must be pre-transaction (AI-based risk scoring) not post-transaction (dispute/chargeback resolution). This changes the value proposition entirely for fraud vendors — from dispute management to pre-authorization AI.

API-First and Cloud-Native Infrastructure

The shift from on-premise core banking to cloud-native API infrastructure (Galileo, Marqeta, Unit, Synctera, Temenos SaaS) is enabling fintechs to launch in months rather than years. This has dramatically lowered barriers to entry at the fintech layer but increased concentration risk at the infrastructure provider layer. Galileo (now SoFi) and Marqeta each power hundreds of fintechs — a Galileo outage disrupts the entire ecosystem.

Payment Orchestration / Smart Routing

Orchestration is the fastest-growing software layer in merchant payments. Merchants with >\$50M in payment volume are increasingly deploying multi-PSP strategies with smart routing based on: authorization rate by BIN/card type/geography, transaction cost, fraud score, latency. Providers: Spreedly (150+ payment connectors), Primer (no-code orchestration), Gr4vy (cloud routing), Checkout.com (orchestration + acquiring), Adyen (unified commerce). **Key strategic insight:** as orchestration commoditizes the acquiring layer, the winners will be the orchestration platforms with the richest decision intelligence (AI-based routing optimization), not the ones with the most connectors.

Tokenization and Network-Linked Wallets

EMV Co (Visa/Mastercard/Amex/Discover consortium) network tokens have replaced primary account numbers (PANs) in mobile wallets and many e-commerce flows. This has a profound strategic implication: tokens are managed by the network (Visa Token Service, Mastercard Digital Enablement Service), which means the networks control lifecycle management of payment credentials even as the consumer-facing experience moves to Apple Pay or Google Pay. Networks become more embedded, not less, in the digital wallet era.

Embedded Finance and Software-Led Distribution

The "embedded payments" thesis: payment functionality built directly into vertical SaaS (Shopify Payments, Toast POS, ServiceTitan field service, FareHarbor activities booking). Merchant retention for embedded payments providers is structurally higher than for standalone processors because switching requires replacing the entire software stack. Take rates are also structurally higher — merchants pay for convenience and integration, not just transaction

processing. This is why Stripe, PayFac-in-a-box providers (Payrix, Finix, Infinicept), and Adyen for Platforms are growing faster than traditional acquirers.

Open Banking / Account Connectivity

CFPB's 1033 final rule (effective 2024–2026 by institution size) mandates financial data portability — consumers can direct banks to share their financial data with third parties via API. This accelerates Plaid, MX Technologies, Akoya, and Finicity (acquired by Mastercard). The strategic implications: (1) reduces Plaid's screen-scraping friction (which creates compliance/security issues); (2) enables richer A2A payment initiation; (3) shifts bank data from captive asset to regulated utility. **Non-obvious second-order effect:** open banking data connectivity + AI analytics = real-time underwriting of creditworthiness without credit bureau dependency. This threatens FICO and traditional credit decisioning.

Stablecoins and Blockchain-Based Settlement

The GENIUS Act framework has legitimized stablecoin infrastructure. The key architectural shift: programmable settlement (smart contracts can automate conditional payments, escrow release, streaming payroll, and DvP for securities) is not possible on ACH or Fedwire. This creates entirely new payment product categories — not just cheaper versions of existing ones.

5. AI DISRUPTION LAYER

Where AI Changes Economics vs. Where It Only Improves Efficiency

This distinction is critical for strategy. **AI changes economics** when it: (a) enables new revenue streams that didn't previously exist; (b) reduces unit costs to near-zero in ways that change pricing power; (c) creates data moats that compound over time. **AI improves efficiency** when it: (a) automates human-labor-intensive tasks without changing the underlying fee structure; (b) reduces fraud losses; (c) speeds up workflows.

Real-Time Fraud Scoring: Economics Changed

Pre-AI fraud detection was rules-based (velocity checks, blacklists, hard thresholds). AI models trained on billions of transactions reduce false positive rates by 30–50% (more revenue for merchants), reduce false negative rates (fewer fraud losses for issuers), and enable dynamic risk scoring that adapts to new fraud patterns in near-real-time. Stripe's AI-driven Radar tool cut card testing attacks by an impressive 80%, analyzing billions of data points in real time. The economic impact is material: lower fraud losses = lower interchange rates required to subsidize losses = structural pressure on current interchange levels. AI fraud tools are also creating a new revenue pool: fraud-as-a-service, where networks, fintechs (Sift, Sardine), and processors charge per-transaction risk scoring fees to issuers and merchants.

AML / KYC / KYB Automation: Efficiency Improvement Now, Economics Shift Later

Legacy AML systems generate ~95% false positive rates on transaction monitoring alerts. AI reduces this to ~70-80%, dramatically cutting the cost of compliance investigations. For large banks spending \$500M-\$2B annually on BSA/AML compliance, this is massive efficiency gain. The economic shift: AI-native compliance vendors (ComplyAdvantage, Hawk AI, Unit21) are disrupting the legacy compliance software market (Oracle FCRM, Actimize/NICE, Temenos Financial Crime Mitigation). Compliance SaaS unit economics improve as AI models scale — a fundamental shift toward recurring SaaS revenue replacing high-labor compliance operations.

Non-obvious insight: as AI automates AML decision-making, banks face a new regulatory risk — model explainability. Regulators (FinCEN, OCC, FRB) require explainable AI for SAR filing and examination purposes. This creates demand for explainable-AI (XAI) tooling in compliance — a new sub-category of compliance software.

Payment Operations Automation: Efficiency Now, Economics Shift Possible

Reconciliation, exceptions handling, return processing, and dispute investigations are among the most labor-intensive payment operations. AI agents (Generative AI + workflow automation) can automate 60-80% of routine exception handling. Modern Treasury, Stripe, and Adyen are already deploying LLM-based payment operations tools internally. The economics shift: if payment operations automation reduces total cost of ownership for merchant payment infrastructure, net margins for PSPs could expand even as transaction fees compress — decoupling revenue from labor cost.

Smart Routing Optimization

AI-based payment routing (not just rule-based routing) optimizes authorization rates, fraud scores, cost, and latency simultaneously across acquirers and payment methods. Providers with richer cross-merchant data can build superior routing models — this creates a **data network effect** where transaction volume generates model improvement that improves authorization rates, which attracts more volume. Primer, Spreedly, and Adyen (with its unified platform data) are building toward this. The key insight: this is where AI **changes economics** — better authorization rates directly translate to higher revenue capture for merchants, creating willingness to pay for intelligence-layer products on top of commodity processing.

Customer Support and Merchant Analytics: Efficiency Improvement

LLM-based customer service (Stripe Sigma, Adyen Analytics, PayPal's AI checkout assistant) reduces support costs and improves merchant experience. This is efficiency, not an economic model shift. The exception: AI-powered merchant analytics that generate actionable revenue insights (e.g., "your authorization rate in Germany drops 15% on Amex — here's what to do") have upsell value and can support premium SaaS tiers.

Transition to Autonomous Decisioning

The medium-term trajectory: **fully autonomous AI agents for payment operations**. This means AI systems that (a) identify exceptions, (b) investigate root causes by querying data sources, (c) remediate issues (re-route, re-submit, flag for human review) without human intervention. Early implementations are live at Stripe, Adyen, and large bank treasury operations. This trajectory creates existential risk for payment operations BPO firms (Conduent, Genpact payment processing) and cost reduction opportunity for any firm currently paying for manual exception handling.

6. COMPETITIVE DYNAMICS AND SHIFTING CONTROL POINTS

How Control Is Shifting Across the Stack

The traditional control point was the issuing bank — it owned the consumer relationship, the credit product, and the payment credential. This control point is fragmenting. Wallets (Apple Pay) now own the consumer checkout experience. Platforms (Shopify, Toast) own the merchant relationship. Orchestration layers own the routing logic. AI vendors own the risk decisioning. The bank retains the deposit relationship and credit extension — which remain powerful — but the intermediary economics at each layer are being competed down.

The key shift: control is moving toward players with superior data and intelligence capabilities. Visa and Mastercard retain control because they see the full transaction (both sides) and can monetize that data through value-added services. Stripe retains control because its fraud and analytics models are trained on aggregate multi-merchant data. The players losing control: commodity acquirers and processors who have transaction volume but not the intelligence layer to monetize it.

Cards vs. A2A: The Real Dynamic

Cards will not be displaced in consumer discretionary payments in the next 5 years. The reasons: (1) rewards economics create consumer lock-in; (2) zero-liability protection has no real-time payment equivalent; (3) credit functionality (revolving balance, installments) is native to cards, not A2A. Where displacement is real and accelerating: (a) bill pay and recurring billers who will shift to A2A/FedNow to capture the lower cost and improve float; (b) B2B AP where commercial card interchange is the most expensive rail; (c) government disbursements (IRS refunds, insurance claims, earned wage access). The consumer debit card is more vulnerable than credit — it has no rewards moat and higher regulatory interchange caps.

Checks and SWIFT: Decline Trajectories

Checks: **structural decline, not disruption**. The volume trajectory is a 6–8% annual decline that will continue. The remaining volume (mostly B2B) will transition to ACH Same-Day and RTP

over the next 5–7 years. There is no scenario where check volume stabilizes.

SWIFT: structural redefinition, not simple decline. ISO 20022 migration is modernizing SWIFT. SWIFT Go is addressing sub-\$10K international transfers. SWIFT itself will survive because it solves the interoperability problem no single bilateral network can. But SWIFT's role as the *only* practical cross-border messaging standard is ending as stablecoin corridors and bilateral API connections emerge. SWIFT becomes one option among several for cross-border.

Embedded Payments and Customer Ownership

Embedded payments fundamentally change pricing power because switching cost is the entire software stack, not just the payment processor. Toast's take rate (~2.49% + flat fee) is 30–50bps higher than a merchant could achieve by buying standalone payment processing — and merchants accept it because the integration value is worth it. This is the embedded finance premium. The implication: payment revenue is becoming SaaS revenue, and the players with vertical SaaS distribution (Shopify, Toast, ServiceTitan, Mindbody) will expand payment take rates even as standalone processor margins compress.

Commoditization vs. Strategic Premium

Commoditizing: Plain-vanilla acquiring (sub-50bps margin), generic gateway functionality, screen-scraping open banking, basic ACH initiation, rules-based fraud tools, legacy core banking.

Gaining strategic premium: AI-based fraud and risk scoring (data network effects), stablecoin issuance/reserve management, payment orchestration with routing intelligence, embedded payments in vertical SaaS, cross-border infrastructure with proprietary local network access, tokenization infrastructure (controlled by networks), and compliance automation with explainable AI for regulated entities.

7. REVENUE POOLS AT RISK

Interchange and Merchant Discount Economics

Pressure sources: Regulatory (Credit Card Competition Act threatens credit card routing choice, which would compress interchange for top-of-wallet premium rewards cards); competition (A2A rails offer merchants 70–90% cost savings on per-transaction basis); AI-driven fraud reduction (reduces the cross-subsidy rationale for high interchange).

Risk level: Medium over 3 years, high over 5 years. The CCCA has been introduced multiple times but not passed — however, the political pressure from merchants is persistent. If CCCA passes, the impact on Visa/Mastercard network fee revenue and issuer interchange could be \$10–20B annually.

Who is most exposed: Premium rewards card issuers (Capital One, Chase Sapphire, Citi Thank You, Amex Platinum). Their business model depends on high interchange subsidizing the rewards liability. If routing choice forces these cards into lowest-interchange routing, the economics collapse.

Where value migrates: A2A processing infrastructure (if volumes shift), and merchant-funded rewards programs (issuers shift reward cost to merchants directly rather than via interchange cross-subsidy).

Network/Scheme Fees

Pressure sources: Visa/Mastercard fee increases have been frozen for 5 years per 2024 settlement. Long-term: CCCA routing competition.

Risk level: Low in near-term (settlement freeze), medium over 5 years.

Mitigation: Networks are rapidly diversifying into value-added services (VAS) — fraud tools, data analytics, A2A connectivity (Visa A2A, Mastercard Send), tokenization services. VAS already ~35% of network revenue and growing. This diversification is both defensive and structurally sound.

Acquiring and Gateway Margins

Pressure sources: Adyen, Stripe, and direct acquirer competition. Orchestration layers abstracting away single-acquirer lock-in. AI improving authorization rates reduces the "value" of acquirer relationships.

Risk level: High and ongoing. This is the most actively commoditizing segment. Gross margin for standalone gateway businesses is approaching nil. Adyen's structural advantage (direct acquiring) is creating 2:1 margin differential vs. ISO-model processors.

Where value migrates: Embedded acquiring in vertical SaaS, payment orchestration with AI routing intelligence, and analytics/underwriting services layered on transaction data.

FX Margins and Cross-Border Spreads

Pressure sources: Wise, Airwallex, and stablecoin rails have already compressed major corridor margins 60–80% vs. bank rates. Stablecoin-based settlement (USDC, RLUSD) eliminates the FX conversion step entirely for USD-to-USD-equivalent flows.

Risk level: High and accelerating. Bank FX cross-border margins are at peak risk. JPMorgan and Citi defend the institutional end (complex multi-currency treasury, trade finance) but lose ground in SMB and consumer corridors.

Who is most exposed: Regional banks with significant cross-border wire income, Western Union (structural secular decline in high-margin consumer remittance corridors).

Where value migrates: Last-mile conversion infrastructure, local payment network access (Nium, Thunes), and stablecoin-to-flat conversion technology.

Wire and SWIFT Fee Income

Pressure sources: FedNow for domestic (lower-cost real-time alternative). Stablecoin rails and Wise/Airwallex alternatives for cross-border.

Risk level: Medium domestic (FedNow volume still small), medium-high international. Wire volume is sticky at the institutional/high-value end. SMB and mid-market international wires are at risk.

Where value migrates: RTP/FedNow transaction fee income (small per-transaction but scalable) and stablecoin reserve float.

Float Income

Pressure sources: Higher interest rates have actually increased float income value in 2023–2025. The risk: rate normalization. Also, CFPB has targeted wallet float (PayPal's FDIC question), and open banking data portability may enable consumers to move balances to higher-yield accounts more easily.

Risk level: Low near-term (rates elevated), medium over 3 years as rates normalize. Stablecoin reserve float (T-bill yield on \$280B+ outstanding) is the fastest-growing float pool — currently very high value.

Fraud Management Margins

Pressure sources: AI is dramatically reducing fraud losses — which erodes the revenue of fraud management services priced as a percentage of losses prevented. As AI drives fraud rates down, the base on which services are priced shrinks.

Paradox: AI simultaneously disrupts fraud service revenue and creates demand for new AI-based fraud services. The transition is from "loss-based pricing" (percentage of fraud prevented) to "SaaS subscription pricing" for AI risk models.

Who is most exposed: Legacy rules-based fraud vendors (older FIS, SAS, Oracle FCRM systems). Who wins: AI-native vendors (Sardine, Featurespace, Sift, Stripe Radar).

AML/KYC/Compliance Service Revenues

Pressure sources: AI automation reduces the human labor in compliance, which is where a significant portion of compliance vendor revenue comes from (implementation, configuration, false-positive investigation workflow). AI also reduces regulatory fines by improving detection quality.

Risk level: Medium. AI-native vendors are displacing legacy systems, but total compliance spending continues to grow due to regulatory scope expansion. The revenue pool is not

shrinking — it is shifting from labor-intensive custom implementations to AI SaaS pricing models.

Where value migrates: Real-time AI model inference at scale (per-API-call pricing, SaaS subscriptions), explainable AI compliance documentation, and automated SAR/CTR reporting.

Correspondent Banking Economics

Pressure sources: Stablecoin rails, Wise/Airwallex bilateral connections, Ripple ODL for specific corridors. De-risking by large correspondent banks reduces coverage, increasing cost concentration.

Risk level: High for SMB and consumer cross-border flows; medium for institutional. The wholesale correspondent banking model for high-value flows (trade finance, securities, interbank) is structurally defensible.

Where value migrates: Local payment network infrastructure, stablecoin issuance float, and treasury-as-a-service for SMBs currently underserved by traditional correspondents.

8. REGULATORY AND POLICY LANDSCAPE

Interchange Regulation

The Durbin Amendment (2010) capped debit interchange for large issuers at $\sim \$0.21 + 0.05\%$. The Federal Reserve proposed expanding Durbin coverage (lower cap) in 2023 — the rule would cut regulated debit interchange by $\sim 30\%$, affecting JPMorgan, BofA, Wells Fargo's debit issuer economics. This has not been finalized as of March 2026 amid legal challenges.

The Credit Card Competition Act (credit card routing mandate) has repeatedly been introduced but not passed. If enacted, it would be the most significant structural change to card economics since Durbin.

CFPB and Consumer Protection

CFPB 1033 open banking rules are in implementation phase, mandating data portability for consumer financial accounts. This is the regulatory catalyst for US open banking — accelerating Plaid, Akoya, Finicity, and bank-direct API access. CFPB has also targeted BNPL (Regulation Z extension for periodic statement and dispute rights) — adding compliance cost to BNPL providers but legitimizing the product category for mainstream adoption.

Stablecoin Regulation (GENIUS Act, July 2025)

The most important payments regulatory development of 2025. The GENIUS Act establishes a regulatory framework for payment stablecoins, prohibiting any person other than a permitted payment stablecoin issuer from issuing a payment stablecoin in the US. Key provisions: 100%

reserve requirement (US dollars and short-term Treasuries), AML/BSA obligations for stablecoin issuers, no interest/yield to holders, state-level regime available for issuers under \$10B outstanding.

Strategic implications: (1) Tether (USDT) cannot legally serve US users without compliance; this could redistribute substantial market share to Circle (USDC) and bank-issued stablecoins. (2) Big Tech companies (Apple, Google, Amazon) cannot issue stablecoins without a GENIUS Act license unless they spin off a compliant subsidiary — the "Stablecoin Certification Review Committee" creates a regulatory hurdle for non-financial Big Tech. (3) Banks that issue stablecoins capture reserve float that currently accrues to Circle — a direct competitive threat to Circle's business model.

AML/BSA Expectations

FinCEN is modernizing BSA rules (AML/CFT Program Effectiveness final rule, 2024) with a risk-based approach that requires actual effectiveness measurement, not just checkbox compliance. This creates demand for AI-based compliance tools that can demonstrate measurable reduction in financial crime risk.

Crypto Regulatory Clarity

The Federal Reserve rescinded its anti-crypto guidance in late 2025 and replaced it with a "different activity, different risks" framework. The OCC granted national trust bank charters to Circle, Paxos, and other nonbank stablecoin issuers in December 2025. This opens the door to federally chartered nonbank stablecoin issuers operating at national scale without state-by-state licensing.

Real-Time Payment Access

The Federal Reserve is considering offering limited "skinny" master accounts to federally regulated stablecoin issuers — giving them access to FedNow and Fedwire without full bank charter requirements. This would be a historic expansion of access to the Federal Reserve payment system to non-bank entities.

9. FUTURE SCENARIOS (2026–2030)

Scenario 1: Base Case — Managed Evolution

Conditions: Regulatory pressure on interchange continues but no CCA passage. FedNow grows to 2,000+ institutions and \$500B+ in annual transaction value. Stablecoins grow to \$500B+ outstanding with GENIUS Act implementation, primarily used in B2B cross-border and institutional flows. Cards retain consumer dominance but face modest share loss in bill pay and B2B.

Winning business models: Visa and Mastercard (network + VAS diversification). Embedded payments in vertical SaaS (Shopify, Toast, Adyen for Platforms). AI-native fraud and compliance vendors (Sardine, Socure, Alloy). Cross-border specialists (Wise, Airwallex). Stablecoin infrastructure (Circle, Fireblocks, bank-issued stablecoins).

Losing business models: Commodity acquirers without software attachment. Legacy AML/KYC systems (rules-based). Correspondent banks in high-volume SMB corridors. Western Union in consumer remittances.

Payment type outcomes: Cards: stable with modest MDR compression. Checks: -7%/year volume. SWIFT: evolves with ISO 20022, adds SWIFT Go, survives. Wallets: continue growing as primary checkout UX, built on card rails. RTP/FedNow: 15–20% of US electronic payment volume by 2028. Stablecoins: \$500B+ outstanding, 2–3% of cross-border B2B volume. Fraud/compliance: 40–50% AI automation of routine operations.

Scenario 2: Accelerated Disruption

Conditions: CCCA passes in 2026–2027, mandating credit card routing choice. FedNow reaches 5,000+ institutions; merchant incentive programs (funded by acquirers seeking to win back volume from cards) drive consumer A2A adoption. Stablecoins reach \$1T+ outstanding; PayPal, Shopify, and several banks launch consumer-facing stablecoin wallets. CFPB data portability rules are aggressively enforced, enabling real-time competitive account switching.

Winning business models: Orchestration platforms with AI routing (Primer, Spreedly). Stablecoin issuers and infrastructure (Circle, Coinbase/Base, bank issuers). FedNow/RTP access middleware (Modern Treasury, Dwolla). AI-native fraud and compliance at scale (data moat advantages). Cross-border challengers (Wise, Airwallex, Nium). Embedded payments SaaS (Shopify Payments, Toast).

Losing business models: Premium rewards card issuers (interchange model broken if CCCA passes). Correspondent banks in cross-border. Legacy payment processors without AI or software differentiation. Rules-based fraud vendors. Standalone gateways.

Payment type outcomes: Cards: significant mix shift from premium rewards to basic/utility; interchange -30–40% on affected volumes if CCCA passes. Checks: accelerates to functional elimination by 2029. SWIFT: significant volume loss in SMB cross-border; institutional flows remain. Wallets: bifurcate into card-backed (current model) and stablecoin-backed (new model). RTP/FedNow: 30%+ of electronic payment volume. Stablecoins: 10–15% of cross-border B2B. Fraud/compliance: 60–70% AI automation; legacy systems face existential pressure.

Scenario 3: Incumbent Resilience

Conditions: CCCA does not pass; regulators maintain status quo on interchange. FedNow

growth slower than expected (consumer protection gap = APP fraud concern limits merchant incentive programs). Stablecoin adoption stalls due to implementation complexity and bank competitive response (tokenized deposits win vs. stablecoins for institutional use). AI improves efficiency at incumbents without changing structure.

Winning business models: JPMorgan (card issuing + institutional infrastructure + tokenized deposits). Visa and Mastercard (ubiquity moat, VAS growth). Large embedded processors (Fiserv/Clover, Global Payments). Marqeta (BaaS infrastructure).

Losing business models: Standalone cross-border fintechs (margin compression without stablecoin tailwinds). Stablecoin-native infrastructure players without bank partnerships. Pure-play fraud SaaS (banks build in-house AI).

Payment type outcomes: Cards: dominant and growing. Checks: slow decline but B2B checks persist through 2030. SWIFT: minimal disruption. Wallets: grow but remain card-backed. RTP/FedNow: grows but limited to niche use cases (payroll, gig economy). Stablecoins: institutional only, limited consumer adoption.

10. STRATEGIC CONCLUSIONS

Where the Market Is Going

The US payments market is transitioning from a **vertically integrated, rail-centric structure** to a **horizontally layered, intelligence-centric structure**. The rails (card networks, ACH, FedNow, Fedwire) are becoming utilities — high volume, low margin, regulated. The intelligence layers (routing optimization, fraud AI, compliance automation, analytics) are becoming the control points and strategic premium zones.

Layers Becoming Utilities

- Plain-vanilla ACH and wire initiation
- Generic payment gateway functionality
- Basic card issuing infrastructure
- Rules-based fraud detection
- Screen-scraping open banking connectivity
- Correspondent account maintenance for standard currencies
- Legacy core banking transaction processing

Layers Becoming Control Points

- **AI-based fraud and risk scoring** (data network effects create self-reinforcing moats)

- **Payment orchestration with routing intelligence** (controls merchant economics)
- **Tokenization infrastructure** (Visa/Mastercard retain strategic premium here)
- **Stablecoin reserve management and issuance** (float income + programmable settlement)
- **Embedded payments in vertical SaaS** (switching cost = entire software stack)
- **Consumer identity and checkout UX** (Apple Pay, Shopify — they own the first click)
- **Compliance AI and explainable decisioning** (regulatory requirement creates demand)
- **Cross-border local payment network access** (scarce and hard to replicate)

Players Best Positioned to Win

1. **Visa and Mastercard:** Network ubiquity + tokenization control + VAS diversification. Even in the disruption scenario, they adapt rather than lose — Visa A2A, Mastercard Send, stablecoin integration are defensive and offensive.
2. **Stripe:** Developer distribution + AI fraud moat + expanding financial services. The most likely enterprise-software-era payments champion.
3. **JPMorgan:** Issuing + acquiring + institutional cross-border + tokenized deposits + AI investment at scale. The bank with the most complete position across the value chain.
4. **Circle (with GENIUS Act legitimacy):** Stablecoin float + USDC institutional trust + bank charter (OCC). Primary beneficiary of regulatory clarity.
5. **Airwallex / Wise:** Cross-border specialists with proprietary local network infrastructure. Hard to replicate and well-monetized.
6. **Adyen:** Direct acquiring + unified commerce + platform payments. The anti-legacy processor.
7. **AI-native compliance vendors (Sardine, Alloy, Socure):** Structural tailwind from AI adoption in compliance at every institution.

Defensible vs. Vulnerable Incumbent Revenue Streams

Most Defensible:

- Card network assessment fees (ubiquity + tokenization moat + no regulatory cap yet)
- Large bank wholesale payments (CHIPS, high-value institutional Fedwire)
- Issuer interest income on revolving credit (not a payment fee but attached to card economics)
- Amex closed-loop data advantage
- JPMorgan/Citi institutional FX for complex multi-currency treasury management

Most Vulnerable:

- Bank FX spread on SMB/consumer cross-border payments (50-80% exposed to Wise/Airwallex/stablecoin over 5 years)
- Check processing revenues (structural elimination)
- Correspondent banking in standard corridors
- Legacy fraud/AML system license revenues (rules-based tools lose to AI)
- Acquiring margins without software attachment

Strategic Moves to Make Now

Banks: (1) Issue stablecoins under GENIUS Act — capture reserve float before Circle dominates. (2) Invest in FedNow/RTP product development for B2B disbursements — capture the volume shift from checks/wires. (3) Build or acquire AI-native compliance infrastructure — or face existential cost disadvantage within 5 years. (4) Compete on embedded finance for SMB with API-first BaaS capabilities.

Card Networks (Visa, Mastercard): (1) Accelerate stablecoin integration (both companies already moving — Visa stablecoin-linked cards, Mastercard stablecoin infrastructure). (2) Defend tokenization control point — own the credential lifecycle regardless of the underlying rail. (3) Build out A2A product suite (Visa A2A, Mastercard Open Banking) to own the A2A rail before being disintermediated by it.

Processors and PSPs (FIS, Fiserv, Global Payments, Adyen, Stripe): (1) Move up the stack into software and intelligence — commodity processing alone is a commoditizing business. (2) Invest in orchestration intelligence and AI routing optimization. (3) Build vertical SaaS or acquire embedded payments software companies.

Fintechs (Stripe, Adyen, PayPal, Block): (1) Accelerate AI model development using proprietary transaction data — this is the defensible moat as payment processing commoditizes. (2) Expand into stablecoin rails for cross-border. (3) Build out financial services beyond payments (lending, treasury, insurance) to increase revenue per customer and switching cost.

Cross-Border Specialists (Wise, Airwallex, Nium): (1) Expand local payment network coverage — this is the hardest-to-replicate moat. (2) Build stablecoin-to-fiat conversion capabilities to serve the growing USDC/stablecoin payment corridor. (3) Move from pure-play FX/payments to full treasury infrastructure for SMB.

AI/Compliance Fintechs (Sardine, Alloy, Socure, ComplyAdvantage): (1) Build out model explainability for regulatory purposes — this is the gate to enterprise bank sales. (2) Expand beyond KYC/fraud to full transaction monitoring and AML workflow automation. (3) Accumulate cross-industry training data to widen the moat against bank in-house AI buildout.

This analysis reflects market conditions and regulatory developments as of Q1 2026. The pace of change in this sector — particularly around stablecoin regulation (GENIUS Act implementation), FedNow adoption, and AI capability advancement — may materially alter specific forecasts within the 3-5 year horizon.